

PRACTICE EXAM 2 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 2

Twenty new questions. Three are deliberate near-twins of Exam 1.

QUESTIONS 20 PAUSE 9 seconds NEW TOPICS 17 of 20

Before you start

- ◆ **Only take this if you have already worked your weakest domain** from exam one. Otherwise you are measuring the same gap twice.
- ◆ Write your answers down again, and score by domain.
- ◆ Resist recognising a question shape. Three of these look familiar and end differently.
- ◆ **Answer every question.** Guessing is still free.
- ◆ The scoring guide compares this paper with exam one, so keep that score to hand.

ON THE EXAM

If your exam one score was below twelve and you have not rewatched anything, stop and do that first.

A company runs a database on Amazon RDS for PostgreSQL. A security audit asks who applies patches to the database engine, who applies patches to the underlying operating system, and who creates and manages database user accounts.

The company wants a clear answer for its auditor.

Which statement is correct?

- ☐ A AWS patches the engine and the OS; the customer manages database users
- ☐ B The customer patches the engine and the OS; AWS manages database users
- ☐ C AWS is responsible for all three
- ☐ D The customer is responsible for all three

◆ PAUSE AND DECIDE

Which statement is correct?

- A

 AWS patches the engine and the OS; the customer manages database users
RDS is managed, so AWS handles both patching layers. Access control never transfers to AWS.
- B

 The customer patches the engine and the OS; AWS manages database users
That describes a database you installed yourself on EC2, and AWS never manages your users.
- C

 AWS is responsible for all three
Who can read your data is always yours, on every service without exception.
- D

 The customer is responsible for all three
True for a self-managed database on EC2. The whole point of RDS is that it is not.

THE SIGNAL

Compare with Exam 1 question 3. Same question shape, different service, and the patching answer moves.

A company runs a document management application on fifteen Linux EC2 instances spread across three Availability Zones. Every instance must read and write the same set of files, and the application expects a mounted filesystem path.

The application source code is not available and cannot be changed.

Which storage service should the company use?

- | | |
|---------------------------------------|---|
| ☐ A Amazon S3 | ☐ B Amazon EBS |
| ☐ C Amazon EFS | ☐ D Amazon FSx for Windows File Server |

◆ PAUSE AND DECIDE

Which storage service should the company use?

- A

 Amazon S3
Object storage over HTTP. It cannot be mounted, so the application would need rewriting.
- B

 Amazon EBS
Attaches to one instance in one Availability Zone. It cannot be shared across fifteen instances in three zones.
- C

 Amazon EFS
A mountable NFS filesystem shared by many Linux instances, spanning Availability Zones.
- D

 Amazon FSx for Windows File Server
Speaks SMB for Windows workloads. These instances run Linux.

THE SIGNAL

Mounted filesystem plus cannot be changed eliminates S3. Linux plus many instances then names EFS.

A retailer used to take eleven weeks to launch a new regional storefront, most of it spent procuring and installing servers. Two of the last four storefronts were closed within a year of opening.

On AWS the same launch now takes nine days, and the retailer trials new regions far more readily than before.

Which benefit of the AWS Cloud does this describe?

- | | |
|---|--|
| ☐ A Elasticity | ☐ B Agility |
| ☐ C Economies of scale | ☐ D High availability |

◆ PAUSE AND DECIDE

Which benefit of the AWS Cloud does this describe?

- A

 Elasticity
Nothing scales with demand. The workload size never changes, only how quickly it can start.
- B

 Agility
Eleven weeks became nine days, and the retailer now experiments more freely. That is speed to act.
- C

 Economies of scale
About aggregated demand lowering unit prices. No pricing mechanism appears here.
- D

 High availability
Availability is about staying up. Nothing here was ever down.

THE SIGNAL

The second of the three paired questions. Exam 1 question 2 changed capacity; this one changes time.

A mobile game needs to store player inventories, retrieved by player identifier. The developers expect anywhere between ten thousand and five million players, cannot predict the growth curve, and need consistent single-digit millisecond reads.

There are no joins, and the team does not want to size or manage database servers.

Which database should the company use?

- | | |
|---|--|
| ☐ A Amazon RDS for MySQL | ☐ B Amazon Redshift |
| ☐ C Amazon DynamoDB | ☐ D Amazon Neptune |

◆ PAUSE AND DECIDE

Which database should the company use?

- A

 Amazon RDS for MySQL
Relational strengths wasted with no joins, and you still choose an instance size and plan capacity.
- B

 Amazon Redshift
An analytical warehouse. Wrong for serving an application in real time.
- C

 Amazon DynamoDB
Serverless, no capacity planning, single-digit millisecond reads at any scale, retrieved by key.
- D

 Amazon Neptune
A graph database. Nothing here traverses relationships between players.

THE SIGNAL

Unpredictable scale, single-digit milliseconds, no servers, retrieval by key. Four DynamoDB signals together.

An application on an EC2 instance needs to write objects to an Amazon S3 bucket. A developer has created an IAM user, generated access keys, and stored them in a file on the instance so the application can read them.

A security review asks for a better approach.

What should the company do?

- ☐ A Encrypt the file containing the access keys
- ☐ B Rotate the access keys every ninety days
- ☐ C Attach an IAM role to the EC2 instance
- ☐ D Store the access keys in AWS Secrets Manager

◆ PAUSE AND DECIDE

What should the company do?

- A

 Encrypt the file containing the access keys
The application must decrypt it to use it, so the key material is still reachable on the instance.
- B

 Rotate the access keys every ninety days
Good hygiene, but a long-term credential still sits on the instance between rotations.
- C

 Attach an IAM role to the EC2 instance
Temporary credentials delivered and rotated automatically. There is no long-term key to leak at all.
- D

 Store the access keys in AWS Secrets Manager
Better than a plain file, but the instance still needs credentials to reach Secrets Manager. It moves the problem.

THE SIGNAL

Three options manage the flaw better; one removes it. An AWS service needing permissions is always a role.

A research institute processes genome sequences overnight. Each job runs for roughly two hours, the jobs are independent, and any job interrupted partway through can simply be started again from the beginning.

Results are needed by nine the next morning, and cost is the primary concern.

Which purchase option is most appropriate?

- | | |
|--|--|
| ☐ A On-Demand Instances | ☐ B Spot Instances |
| ☐ C Three-year Reserved Instances | ☐ D Dedicated Hosts |

◆ PAUSE AND DECIDE

Which purchase option is most appropriate?

- A

 On-Demand Instances
No commitment needed, but far dearer than Spot for work that explicitly tolerates interruption.
- B

 Spot Instances
Independent, restartable jobs with a generous deadline. All three Spot conditions are stated.
- C

 Three-year Reserved Instances
Commits to steady capacity for work that runs only overnight, in varying volume.
- D

 Dedicated Hosts
Dedicated hardware at a premium, with no licensing or compliance requirement in sight.

THE SIGNAL

Independent, restartable, and a deadline hours away. Spot needs all three, and the scenario states all three.

A company runs its platform in a single Region across three Availability Zones. Availability has been excellent and there has been no outage in two years. Users in Europe and North America are satisfied.

Users in Southeast Asia report that pages take four to six seconds to load. The content they are requesting is largely static and identical for every user.

What should the company do?

- ☐ A Deploy a full disaster recovery environment in a second Region
- ☐ B Add a fourth Availability Zone
- ☐ C Use Amazon CloudFront
- ☐ D Move to larger instance types

◆ PAUSE AND DECIDE

What should the company do?

- A

 Deploy a full disaster recovery environment in a second Region
Solves Region loss, which is not the problem. Nothing has been unavailable in two years.
- B

 Add a fourth Availability Zone
More availability inside the same Region. All the zones are equally far from Southeast Asia.
- C

 Use Amazon CloudFront
Caches identical static content at edge locations near those users, so it is served locally.
- D

 Move to larger instance types
Users elsewhere are satisfied, so the application is not slow. The distance is.

THE SIGNAL

The third paired question. Exam 1 question 18 named a Region failure; this one names distance.

A team ships one large release each month, has no automated rollback, and diagnoses incidents by reading logs on individual servers. It proposes to deploy small changes frequently, script its rollback, centralise its logging, and hold a review after every incident. No change to the application architecture is planned.

Which Well-Architected pillar do these changes belong to?

- | | |
|---|---|
| ☐ A Reliability | ☐ B Operational Excellence |
| ☐ C Performance Efficiency | ☐ D Security |

◆ PAUSE AND DECIDE

Which Well-Architected pillar do these changes belong to?

- A

 Reliability
Tempting, since fewer incidents means more uptime. But Reliability lives in architecture, and none is changing.
- B

 Operational Excellence
Small frequent changes, scripted rollback, observability and learning from incidents are its design principles.
- C

 Performance Efficiency
No question of resource selection or efficiency appears anywhere.
- D

 Security
Centralised logs support security, but the stated purpose is diagnosing deployments.

THE SIGNAL

Every proposed change is a process change, and the last sentence confirms the architecture is untouched.

A company's security team is alerted that an EC2 instance is making outbound connections to an address associated with a known botnet, and that an IAM access key belonging to a developer is being used from a country where the company has no offices.

The team wants a service that would surface both automatically, with nothing installed on the instances.

Which service should the company enable?

- | | |
|---|---|
| ☐ A Amazon Inspector | ☐ B Amazon GuardDuty |
| ☐ C AWS Config | ☐ D Amazon Macie |

◆ PAUSE AND DECIDE

Which service should the company enable?

- A

 Amazon Inspector
Finds vulnerabilities that could be exploited. Neither symptom here is a missing patch.
- B

 Amazon GuardDuty
Malicious network destinations and credential use from an improbable location are core findings. Agentless.
- C

 AWS Config
Records configuration and evaluates rules. It sees neither network behaviour nor credential use.
- D

 Amazon Macie
Looks inside S3 for sensitive data. No data content is involved in either symptom.

THE SIGNAL

Two behaviours, no vulnerability and no configuration change. Nothing to install is a GuardDuty hallmark.

An Amazon S3 bucket's policy was changed at some point overnight, briefly allowing public access. The company needs to establish which IAM identity made the change and at what time.

It has CloudTrail, CloudWatch and Config all enabled.

Which service answers this question?

- | | |
|--|--|
| ☐ A Amazon CloudWatch | ☐ B AWS Config |
| ☐ C AWS CloudTrail | ☐ D AWS Trusted Advisor |

◆ PAUSE AND DECIDE

Which service answers this question?

- A

 Amazon CloudWatch
Metrics, logs and alarms about performance. It does not record who called an API.
- B

 AWS Config
Would show that the bucket was public and for how long, but it does not name the caller.
- C

 AWS CloudTrail
Records the API call, the identity behind it and the timestamp. The who question is always CloudTrail.
- D

 AWS Trusted Advisor
Flags public buckets as a current finding. It keeps no history and names no identity.

THE SIGNAL

Which identity and at what time. Both are event questions, and events with an actor are always CloudTrail.

A company runs an ageing self-hosted customer relationship management system. Rather than move it, the company has decided to cancel it entirely and subscribe to a commercial CRM product delivered as software as a service.

No code will be migrated and the company will no longer run the software itself.

Which migration strategy is this?

- | | |
|---------------------------------------|-------------------------------------|
| ☐ A Replatform | ☐ B Refactor |
| ☐ C Repurchase | ☐ D Retain |

◆ PAUSE AND DECIDE

Which migration strategy is this?

- A

Replatform
Keeps your software and swaps one component for a managed service. Here the software is abandoned entirely.
- B

Refactor
Rewriting the system yourself. You would still own the software afterwards; here you do not.
- C

Repurchase
Drop and shop. The company stops owning software and subscribes to a product instead.
- D

Retain
Deliberately leaving something where it is. This system is being replaced, not kept.

THE SIGNAL

Who owns the software afterwards decides it. Refactor keeps it and reshapes it; repurchase gives it up.

A bank must store the keys protecting its most sensitive records in hardware that no other AWS customer shares. Its regulator requires the bank to demonstrate that AWS personnel cannot access the key material under any circumstances. Cost and operational effort are secondary.

Which service meets this requirement?

- ☐ A AWS Key Management Service with customer managed keys
- ☐ B AWS CloudHSM
- ☐ C AWS Secrets Manager
- ☐ D AWS Certificate Manager

◆ PAUSE AND DECIDE

Which service meets this requirement?

- A

AWS KMS with customer managed keys
The near miss. You control the key policy, but the underlying hardware is managed and multi-tenant.
- B

AWS CloudHSM
Single-tenant hardware security modules under the bank's exclusive control.
- C

AWS Secrets Manager
Stores and rotates credentials. It is not a hardware key store.
- D

AWS Certificate Manager
Issues and renews TLS certificates. Unrelated to storing encryption keys in hardware.

THE SIGNAL

No other customer shares the hardware, and AWS staff cannot reach the keys. Only CloudHSM satisfies both.

An order service writes an order, then calls a stock service, a payment service and a notification service one after another. When the notification service is slow, order submissions time out and customers give up.

The company wants order submission to return immediately, with the downstream work guaranteed to happen afterwards.

Which service should the company introduce?

- | | |
|---|---|
| ☐ A Amazon SNS | ☐ B Amazon SQS |
| ☐ C Amazon EventBridge | ☐ D AWS Step Functions |

◆ PAUSE AND DECIDE

Which service should the company introduce?

A Amazon SNS

Pushes to subscribers but does not hold work. A slow or unavailable subscriber is still a problem.

C Amazon EventBridge

Routes events to targets by rule. It does not buffer work when a consumer is slow.

B Amazon SQS

The order service writes to a queue and returns at once. Messages persist until a worker processes them.

D AWS Step Functions

Orchestrates ordered steps. The problem here is coupling to a slow consumer, not sequencing.

THE SIGNAL

Return immediately plus guaranteed to happen afterwards describes buffering with persistence. That is a queue.

A finance director wants a report showing last quarter's AWS spend broken down by each of the company's six departments. Each department has its own AWS account under one organisation.

Separately, the director wants an email as soon as any department's forecast spend indicates it will exceed its allocation this month.

Which two services meet these requirements? (Choose two.)

- ☐ A AWS Cost Explorer
- ☐ B AWS Budgets
- ☐ C AWS Pricing Calculator
- ☐ D AWS Cost Anomaly Detection
- ☐ E AWS Trusted Advisor

◆ PAUSE AND DECIDE

Which two services meet these requirements? (Choose two.)

- A

AWS Cost Explorer
Analyses historical cost grouped by linked account. Exactly the departmental breakdown.
- B

AWS Budgets
Sets a threshold per account and alerts on forecast spend crossing it.
- C

AWS Pricing Calculator
Estimates costs for things not yet built. It never sees actual spend.
- D

AWS Cost Anomaly Detection
Alerts on unusual spikes, not on crossing a planned allocation. A steady overspend never triggers it.
- E

AWS Trusted Advisor
Best-practice checks including idle resources. It produces no departmental spend report.

THE SIGNAL

Backward-looking breakdown is Cost Explorer. Forward-looking threshold alert is Budgets. One tool each.

A migration has stalled. The infrastructure is built and working, but the operations team has never run a cloud environment, is anxious about losing its expertise, and no training programme has been arranged.

The programme director wants to know which AWS Cloud Adoption Framework perspective this belongs to.

Which perspective is it?

- | | |
|---------------------------------------|-------------------------------------|
| ☐ A Operations | ☐ B People |
| ☐ C Platform | ☐ D Business |

◆ PAUSE AND DECIDE

Which perspective is it?

- A

 Operations
The bait. The stem names the operations team, but Operations covers monitoring, incidents and service levels.
- B

 People
Missing skills, anxiety about change, and no training programme. All three are People capabilities.
- C

 Platform
Building the environment, which the stem says is already built and working.
- D

 Business
Aligning investment with objectives. No business case concern appears.

THE SIGNAL

A named team in the stem is not a named perspective. Read the gap, which here is skills and attitude.

A company with ninety AWS accounts requires that no account outside its research organizational unit can create resources in Regions beyond the European Union, whoever attempts it. Several accounts have local administrators holding full IAM permissions. The restriction must not be removable by those administrators.

How should the company enforce this?

- ☐ A An IAM policy denying other Regions, attached to every user
- ☐ B A service control policy on the relevant organizational units
- ☐ C An AWS Config rule flagging resources in other Regions
- ☐ D An IAM permissions boundary on each administrator

◆ PAUSE AND DECIDE

How should the company enforce this?

- A

 An IAM policy denying other Regions, attached to every user
An administrator with full IAM permissions can detach it. It does not survive the stated conditions.
- B

 A service control policy on the relevant organizational units
Sets a ceiling above IAM that no identity in those accounts can exceed, including local administrators and root.
- C

 An AWS Config rule flagging resources in other Regions
Detects the violation after it happens. The requirement is that it cannot happen.
- D

 An IAM permissions boundary on each administrator
Still an IAM construct that an administrator with full IAM permissions could alter.

THE SIGNAL

Whoever attempts it, and not removable by administrators. Only a control above IAM satisfies both.

A company runs a multiplayer game server that communicates over UDP and requires extremely low latency. Its clients are configured with fixed IP addresses that cannot be changed without a client update.

It needs to distribute connections across many backend instances.

Which load balancer should the company use?

- | | |
|--|--|
| ☐ A Application Load Balancer | ☐ B Network Load Balancer |
| ☐ C Gateway Load Balancer | ☐ D Classic Load Balancer |

◆ PAUSE AND DECIDE

Which load balancer should the company use?

- A

 Application Load Balancer
Handles HTTP and HTTPS only. It cannot carry UDP at all, and it provides no static IP.
- B

 Network Load Balancer
Layer 4 with UDP support, extreme performance, and a static IP address per Availability Zone.
- C

 Gateway Load Balancer
For deploying third-party virtual appliances such as firewalls. This is not an appliance.
- D

 Classic Load Balancer
Previous generation. Still supported in a VPC, but never the right answer for a new design.

THE SIGNAL

UDP, extreme low latency and fixed IP addresses. Three separate NLB signatures in one requirement.

A finance manager asks why AWS can offer lower prices than the company achieves running its own data centre, given that AWS also has to buy hardware, power and networking.

An architect explains that AWS combines the usage of very large numbers of customers when it purchases.

Which advantage of cloud computing is the architect describing?

- | | |
|---|--|
| ☐ A Trade capital expense for variable expense | ☐ B Benefit from massive economies of scale |
| ☐ C Stop guessing capacity | ☐ D Go global in minutes |

◆ PAUSE AND DECIDE

Which advantage of cloud computing is the architect describing?

- A

 Trade capital expense for variable expense
About the shape of your payments. The question is about why AWS's unit costs are lower.
- B

 Benefit from massive economies of scale
Aggregating usage across hundreds of thousands of customers lowers unit cost, and AWS passes it on.
- C

 Stop guessing capacity
About provisioning to measured demand rather than a forecast. No capacity decision appears here.
- D

 Go global in minutes
About reach into new Regions. Nothing geographic is described.

THE SIGNAL

Combines the usage of very large numbers of customers is the definition, almost word for word.

An engineering firm of six hundred staff already runs Microsoft Active Directory on premises. It is opening eighteen AWS accounts, one per project. Staff should sign in to whichever accounts they are entitled to using their existing corporate credentials.

The identity team refuses to maintain separate credentials in each account.

Which service meets these requirements?

- ☐ A Create IAM users in each account
- ☐ B Amazon Cognito
- ☐ C AWS IAM Identity Center
- ☐ D An IAM group in the management account

◆ PAUSE AND DECIDE

Which service meets these requirements?

- A

 Create IAM users in each account
Six hundred staff across eighteen accounts, all needing joiner and leaver handling. Exactly what was refused.
- B

 Amazon Cognito
For your application's end users, not staff signing in to AWS accounts. Wrong population.
- C

 AWS IAM Identity Center
Connects the existing directory and grants access across many accounts from one sign-in.
- D

 An IAM group in the management account
Groups organise users inside one account. They neither span accounts nor connect a directory.

THE SIGNAL

Many accounts, an existing corporate directory, and no duplicate credentials. Three signals, one answer.

A company must run a legacy application that requires an unsupported operating system version, needs administrative access to tune kernel parameters, and must run continuously without interruption.

The company asks which AWS compute option is suitable.

Which service should the company use?

- | | |
|---------------------------------------|--|
| ☐ A AWS Lambda | ☐ B AWS Fargate |
| ☐ C Amazon EC2 | ☐ D AWS Elastic Beanstalk |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

AWS Lambda
No OS access, supported runtimes only, and a fifteen-minute limit. It fails all three requirements.
- B

AWS Fargate
No access to the host or its kernel. Containers share a host kernel that cannot be tuned per task.
- C

Amazon EC2
Full control of the operating system, any OS image, and it runs continuously.
- D

AWS Elastic Beanstalk
Provisions specific managed platforms. An unsupported OS version is outside what it offers.

THE SIGNAL

Kernel parameters and an unsupported OS eliminate managed services by definition. Sometimes EC2 is right.

Scoring guide

DOMAIN	QUESTIONS	IF THIS SCORE IS WORSE THAN EXAM 1
D1 Cloud Concepts	3, 8, 11, 15, 18	You are recognising benefit names rather than reading the scenario.
D2 Security	1, 5, 9, 12, 16, 19	Rewatch module 5 first. Most D2 errors start with the responsibility line.
D3 Technology	2, 4, 7, 10, 13, 17, 20	Check whether the misses share a service family, then rewatch that module only.
D4 Billing	6, 14	Two questions is noise. Compare across all three papers instead.

ON THE EXAM

Questions 1, 3 and 7 were the paired near-twins of Exam 1. If you missed those specifically, you recognised rather than read.

What to do next

- ◆ **Compare domain scores across both papers**, not the totals. A domain weak in both is your real gap.
- ◆ A domain that improved was probably a knowledge gap. One that did not is likely a reading habit.
- ◆ Leave several days before exam three. Recall after a gap is what predicts a pass.
- ◆ Rewatch module 21, the confusion pairs, in that gap. It is built for exactly this point.
- ◆ **If both papers scored above sixteen**, spend the gap on module 22 rather than more revision.

ON THE EXAM

Exam three is the last one. Space it out properly; taking it tomorrow tells you almost nothing new.